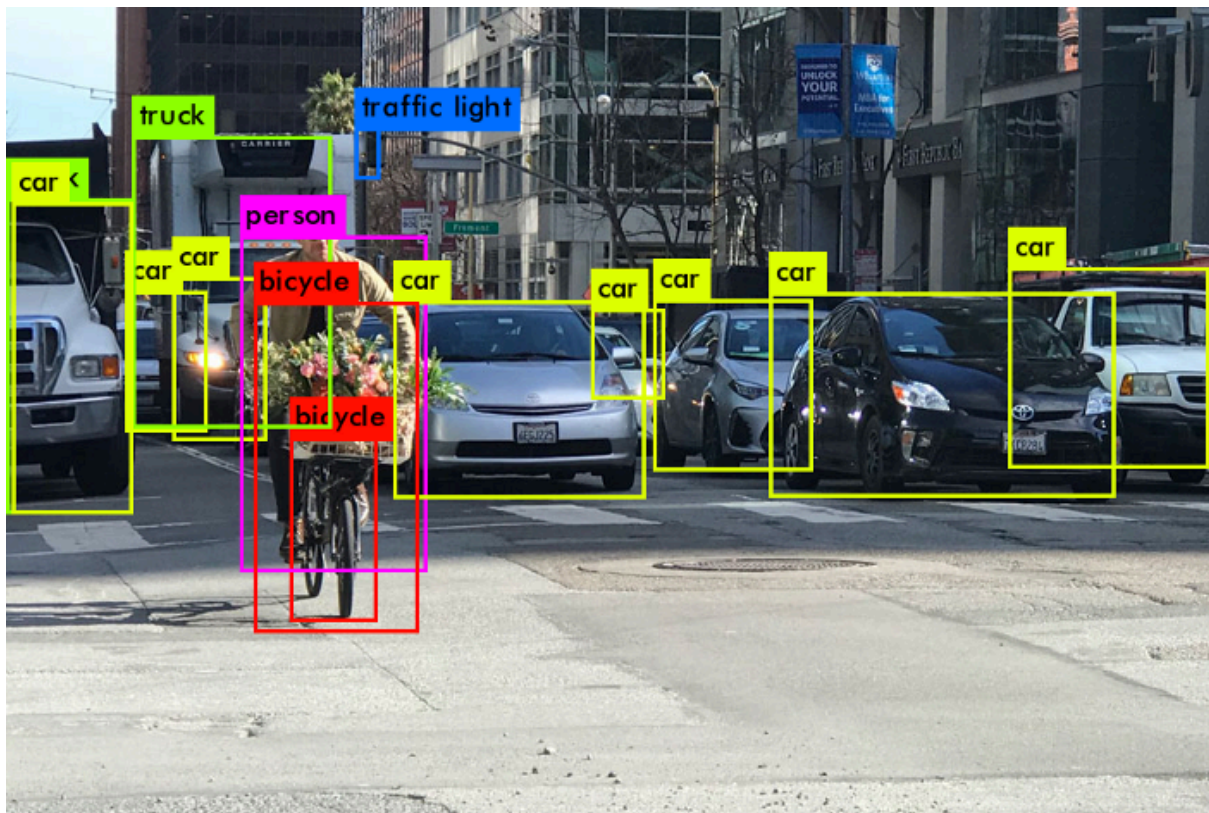


Mono/Stereo Vision-based Alternatives to LiDAR

PROPOSED CHANGE: Determining the optimal vision setup for a Formula Student Car

Aden Trantan

S2 2025 - S2 2026



Background & Motivation

MMS driverless currently relies on the Baraja LiDAR for perception. Whilst this works, it also comes with many drawbacks, including:

- It is expensive
- It is very heavy
- It is very power-hungry
- It does not detect colour
- We aren't even supposed to have the Baraja

However, there is an alternative that could address all of these issues. Cameras are much cheaper, lighter and provide RGB data (which allows us to classify cones and outliers much easier). This would lead to very positive effects on the overall performance of the car: simplifying hardware, lowering vehicle mass, reducing costs and optimizing path planning algorithms (can implement colour cost functions).

I want to explore the possibility of a real-time camera system with a mono or stereo vision approach that is built specifically for the MMS driverless problem (Formula Student Driverless constraints), to replace the current cone detection methods that use LiDAR in the perception stack.

My FYP gives me a great opportunity to contribute to MMS research and development, as well as work on something I am very interested in learning more about. Since I'm doing a double degree, I don't have space for any elective units, so I can't take the computer vision unit. I believe this hands-on approach will be the best way for me to learn and understand the ideas, processes and implementation of camera systems and computer vision whilst being able to improve the performance of Monash Motorsport for future years.

Scope

I will design, implement and evaluate a real-time computer vision pipeline to detect and localize cones from a camera input. The final system will be benchmarked against the current LiDAR setup and should demonstrate its potential as a feasible and beneficial replacement for the LiDAR.

Detecting cones:

- Use a lightweight neural network model to detect cones in RGB images (supervised learning with a set of data, **Bryce's data?**)
- Use colour information to assign classifications for cones
- Ensure that it works in real time when connected to the Orin
- Test cone detection with different environmental situations (where colour may be modified/obstructed in some way)

Depth estimation:

- Stereo vision: use disparity from a dual camera setup
- Monocular vision: use deep learning models (supervised/unsupervised learning)
- Make the decision on whether stereo or mono vision will be more beneficial (based on results vs LiDAR performance)
- Convert detected cones into vector data with accurate pose relative to the car
- Output a set of 3D coordinates accurately representing the real-world position of the cones from the camera frame

Comparison and discussion

- Compare the results with the current LiDAR performance
- Test on Formula Student tracks
- Quantify tradeoffs

Integration with MMS

- Ensure the system is compatible with ROS2, MMS WS2
- Provide documentation and tests for future team use

Prior Work

Cameras Research Report (2018) - Mithin Hunsur, Matthew Lane

 M19-D Research Report - Cameras.pptx

 Camera CR.pptx

 Cameras PDR

 Mithun Hunsur - Camera-based mapping and localisation for an autonomous racing ve...

- Lots of information that can be used as a reference/general overview
- It's probably very outdated by now
- Relevance: Still a good set of resources to refer to

Perception Research Report (2020) - Christopher Adis, Bryce Ferenczi

 Perception RR 2020 M21

- Used a stereoscopic camera setup which runs YOLOv4 object detection model on one camera to identify cones
- Used template matching to find cones in the other camera to estimate disparity
- Estimations were generally still too noisy to use in SLAM
- Further research was dropped after the transition from Velodyne to Baraja
- Relevance: Bryce suggests I can train a new detector using their original labelled training data (saves a lot of time) and a more up to date object detection model

Perception Validation Report (2023) - Alex Davies, Eric Wong

 Perception 2023 VR Presentation

 Knowledge Transfer - Pylon C API 2023

 Camera Implementation and Design Report

- Goes into detail about hardware specifics

- Design report timelines the research of cameras for MMS and talks about how it was built up until that point
- Talks about cameras in conjunction with the LiDAR, whereas we want to ditch the LiDAR completely in favour of cameras
- Relevance: Another good set of resources to refer to

Dr Hoam alternative project direction

- Investigate the next generation system
- LiDAR and camera
- Potentially cheaper LiDAR and camera
- Exploring different options
- Part 1: select a direction
- Part 2: implement a baseline solution and compare results with current setup

I need a solid outline of the project objectives and aims by week 7 interview

- Narrow down by then

It **can** be open ended (one solution as the example)

“I want to include that exploration as part of the project”

Look at indycar driverless

Candy and Jed talk

- What is the optimal camera setup?
- What sort of cameras will we use?
- More discussion on the cameras, trying to shift away from LiDAR due to hard constraints
- Mono vs stereo?
- What neural nets do we need to get the distance accurate?
- Orin is enough (built for computer vision)
- How much computing power do we need for a system like this to work?
- WE have an excess of computational power (orin)
- Lidar is a kilo
- 2 camera system is much less than that
- THE CAR IS 250KG, EVERY GRAM COUNTS
 - Will nab us additional points (about 15 points per kilo)
- Justify that the cameras solutions is the best one
- Current velo couldn't detect cones reliably and far away (16 channel vs solid state)
- Much denser point cloud, easier to detect cones, no motion blur

Part 2: implementation

Camera AI architectures

Option 1: go with Hoam's idea of broadening the scope and considering a bunch of different options (WHAT WILL WORK FOR MMS AND WHY? Perception)

Option 2: exploring IF we had a camera-based system, what's the best way we can implement it with SLAM? What algorithms? (State-estimation and perception related decision), with Mehrtash

- **Neural net sensor models?**
- **Mackenzie Yandel FYP**